

"99.9% of science is boring."
- Makise Kurisu

PRECALCULUS

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1 FUNCTIONS

1.1 Intro To Functions

Definition 1.1 (Function). A function is a relationship that maps 1 input to 1 output.

Note: An input never gives more than 1 output.

A function usually looks like

$$y = f(x)$$

x is the *independent* variable.

y or $f(x)$ is the *Dependent* variable.

Where the standard ordered pairs looks like

$$(x, y) \vee (x, f(x))$$

Definition 1.2 (Domain). A domain is a set of inputs.

Definition 1.3 (Range). A range is a set of outputs.

Example 1.1 (Real Example).

Work Hours	Pay
25	\$500
53	\$310
30	\$490
40	\$490

Note: This is a function but not a *one-to-one function*.

Domain : 23, 53, 30, 40

Range : \$500, \$310, \$490

Example 1.2 (Input Output).

$$S = (-2, 16), (-1, 4), (0, 3), (1, 4)$$

This is a function.

$$R = (-2, 5), (3, 9), (5, 0), (-2, 6)$$

This is not a function.

Example 1.3 (Algebraic).

$$y = \frac{1}{2}x - 2$$

This is a function.

$$y = 2x^2 - 5x + 4$$

This is a function.

$$y = \pm\sqrt{3-2x}$$

This is not a function because we get more than 1 output for only 1 input.

Note: $y^2 = x \neq y = \sqrt{x}$, instead it equals to $y = \pm\sqrt{x}$. Which makes it not a function.

Example 1.4 (Factoring).

$$xy + 2y = 3x - 1$$

$$y(x + 2) = 3x - 1$$

$$y = \frac{3x - 1}{x + 2}$$

1.2 Evaluating Functions

Note:

$$y = -7x + 5 \equiv f(x) - 7x + 5$$

$$y = \frac{3}{2}x - 1 \equiv g(x) = \frac{3}{2}x - 1$$

We use the $f(x)$ notation to know what the input is.

Example 1.5 (Exponents).

$$h(x) = -2x^2 + x - 1$$

1. $h(-1) = -2(-1)^2 + (-1) - 1$
 $h(-1) = -4$
 $(-1, -4)$
2. $h(-x) = -2(-x)^2 + (-x) - 1$
 $h(-x) = -2x^2 + x - 1$
3. $h(x+1) = -2(x+1)^2 + (x+1) - 1$
 $h(x+1) = -2(x^2 + 2x + 1) + (x+1) - 1$
 $h(x+1) = -2x^2 - 4x - 2 + x + 1 - 1$
 $h(x+1) = -2x^2 - 3x - 1$

Example 1.6 (Square Root).

$$f(x) = \sqrt{x^2 - 3x}$$

1. $f(5) = \sqrt{(5)^2 - 3(5)}$
 $f(5) = \sqrt{10}$
 $(5, \sqrt{10})$
2. $f(0) = \sqrt{(0)^2 - 3(0)}$
 $f(0) = \sqrt{0}$
 $(0, 0)$
3. $f(2x) = \sqrt{(2x)^2 - 3(2x)}$
 $f(2x) = \sqrt{4x^2 - 6x}$
4. $f(x+h) = \sqrt{(x+h)^2 - 3(x+h)}$
 $f(x+h) = \sqrt{x^2 + 2xh + h^2 - 3x - 3h}$
5. $f(1) = \sqrt{(1)^2 - 3(1)}$
 $f(1) = \sqrt{-2}$

Note: This is a domain issue so we can't use 1.

Note: For domains:

1. **Square Roots:** The inside of the square root (*radicand*) must always be positive.
2. **Denominator:** The denominator must never be 0.

Example 1.7 (Fractions).

$$g(x) = \frac{2x+1}{x-5}$$

$$1. \quad g(0) = \frac{2(0)+1}{(0)-5}$$

$$g(0) = \frac{1}{-5}$$

$$2. \quad g(x^2) = \frac{2(x^2)+1}{(x^2)-5}$$

$$g(x^2) = \frac{2x^2+1}{x^2-5}$$

$$3. \quad g(5) = \frac{2(5)+1}{(5)-5}$$

$$g(5) = \frac{11}{0}$$

Note: This is also a domain issue.

$$4. \quad -g(x) = -\frac{2x+1}{x-5}$$

1.2.1 Difference Quotient

Definition 1.4 (Difference Quotient). A formula used in algebra and calculus to find the **average rate of change** of a function over a small interval

$$\frac{f(x+h) - f(x)}{h}$$

Example 1.8 (Example).

$$f(x) = x^2 + 1$$

$$f(x+h) = (x+h)^2 + 1$$

$$f(x+h) = x^2 + xh + xh + h^2 + 1$$

$$f(x+h) = x^2 + 2xh + h^2 + 1$$

$$\frac{f(x+h)-f(x)}{h} = \frac{(x^2+2xh+h^2+1)-(x^2+1)}{h}$$

$$\frac{f(x+h)-f(x)}{h} = \frac{2xh+h^2}{h}$$

$$\frac{f(x+h)-f(x)}{h} = \frac{h(2x+h)}{h}$$

$$\frac{f(x+h)-f(x)}{h} = 2x + h$$

1.3 Finding Domains

Definition 1.5 (Domains). Domains is the set of inputs of a function that give a $\mathbb{R}$ number output.

1.3.1 The Problem Areas

1. **Square Roots:** $\sqrt{(\quad)}$ radicand > 0 .
2. **Denominators:** $\text{denominator} \neq \text{zero}$.

Example 1.9 (Square Root).

$$\begin{aligned} f(x) &= \sqrt{5-4x} \\ 5-4x &\geq 0 \\ -4x &\geq -5 \\ x &\geq \frac{-5}{-4} \\ x &\leq \frac{5}{4} \end{aligned}$$

Can be written as

$$D = \{x \mid x \leq \frac{5}{4}\}$$

Or

$$(-\infty, \frac{5}{4}]$$

Example 1.10 (Denominator).

$$\begin{aligned} g(t) &= \frac{5t}{t^3-16t} \\ t^3-16t &= 0 \\ t(t^2-16) &= 0 \\ t(t-4)(t+4) &= 0 \\ t \neq 0 \vee t \neq 4 \vee t \neq -4 \end{aligned}$$

Can be written as

$$D = \{t \in \mathbb{R} \mid t \neq 0, t \neq -4, t \neq 4\}$$

Or

$$(-\infty, -4) \cup (-4, 0) \cup (0, 4) \cup (4, \infty)$$

Example 1.11 (Examples).

1. $f(x) = 3x^2 - 2x + 4$
 $D : \{x \in \mathbb{R}\} \text{ Or } (-\infty, \infty)$

2. $h(x) = \frac{\sqrt{3x-15}}{x-4}$

$$x \neq 4$$

$$3x - 15 \geq 0$$

$$3x \geq 15$$

$$x \geq 5$$

$$D : \{x \mid x \neq 4\} \text{ or } [5, \infty)$$

3. $g(x) = \frac{5}{\sqrt{x-8}}$

$$x - 8 > 0$$

$$D : \{x \mid x > 8\} \text{ or } (8, \infty)$$

1.4 Operations Of Functions

Definition 1.6 (Operations Of Function). Function operations involve combining two or more functions to create a new function using standard arithmetic: **addition, subtraction, multiplication and division.**

Example 1.12 (Operations).

$$f(x) = \frac{2x+3}{3x-2} \quad g(x) = \frac{4x}{3x-2}$$

$$D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

1.

$$(f+g)(x) = \frac{2x+3}{3x-2} + \frac{4x}{3x-2} \rightarrow \frac{6x+3}{3x-2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

2.

$$(f-g)(x) = \frac{2x+3}{3x-2} - \frac{4x}{3x-2} \rightarrow \frac{-2x+3}{3x-2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

3.

$$(f \cdot g)(x) = \left(\frac{2x+3}{3x-2} \right) \left(\frac{4x}{3x-2} \right) \rightarrow \frac{8x^2+12x}{(3x-2)(3x-2)} \rightarrow \frac{8x^2+12x}{(3x-2)^2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

4.

$$\left(\frac{f}{g} \right)(x) = \frac{\frac{2x+3}{3x-2}}{\frac{4x}{3x-2}} \rightarrow \frac{2x+3}{3x-2} \cdot \frac{3x-2}{4x} \rightarrow \frac{2x+3}{4x} \quad D : \left\{ x \mid x \neq \frac{2}{3}, x \neq 0 \right\}$$

Example 1.13 (Square Root and Fractions).

$$f(x) = \sqrt{x+3} \quad g(x) = \frac{5}{x}$$

$$D : \{x \mid x \geq -3\} \quad D : \{x \mid x \neq 0\}$$

1.

$$(f+g)(x) = \sqrt{x+3} + \frac{5}{x} \quad D : \{x \mid x \geq -3, x \neq 0\}$$

2.

$$(f-g)(x) = \sqrt{x+3} - \frac{5}{x} \quad D : \{x \mid x \geq -3, x \neq 0\}$$

3.

$$(f \cdot g)(x) = \frac{5\sqrt{x+3}}{x} \quad D : \{x \mid x \geq -3, x \neq 0\}$$

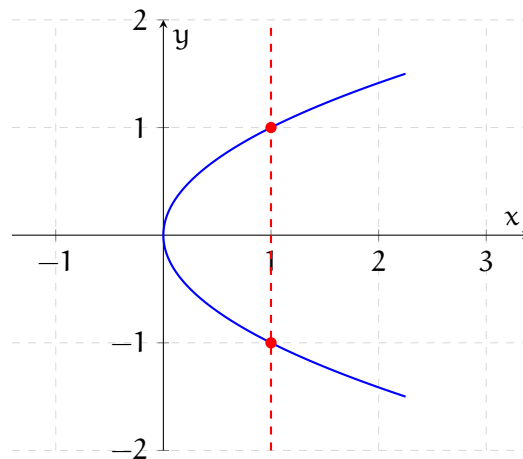
4.

$$\left(\frac{f}{g} \right)(x) = \sqrt{x+3} \cdot \frac{x}{5} \rightarrow \frac{x\sqrt{x+3}}{5} \quad D : \{x \mid x \geq -3, x \neq 0\}$$

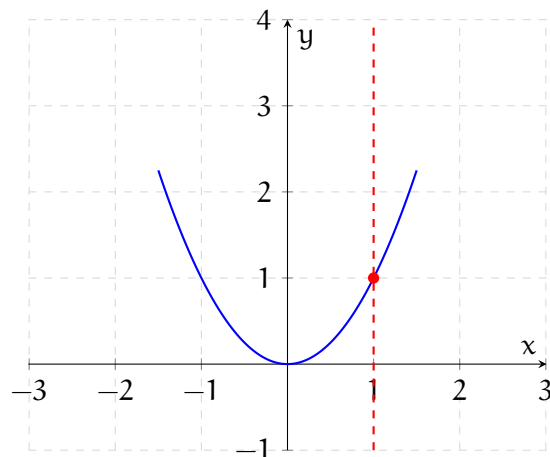
1.5 The Vertical Line Test

Definition 1.7 (Vertical Line Test). This is a test to determine if a graph is a function or not.

Example 1.14 (Non Function). This is not a function because the vertical line passes through 2 points, which means that it outputs more than 1 output.



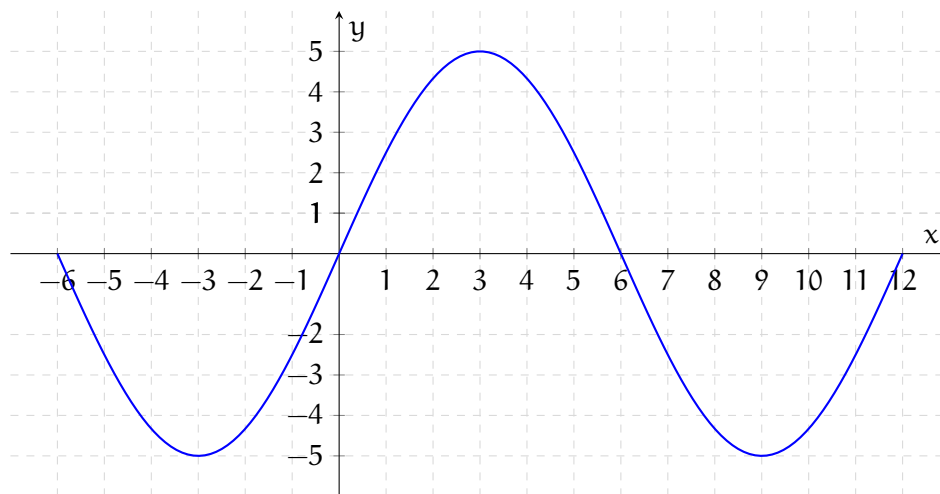
Example 1.15 (Function). This is a function because the vertical line passes through only 1 point, which means that it only gives a single output.



1.6 Features of Graphs Points, Domain, Range

Definition 1.8 (Graph Points). A graph point of a function is an ordered pair (x, y) on a coordinate plane, where y is the output generated when the function precesses the input x .

Example 1.16 (Graphs).



1. Find $f(0) = 0$ $(0, 0)$
2. Find $f(-2) = -2$ $(-2, -2)$
3. Is $f(1)$ positive $\vee$ negative = positive
4. Is $f(-2)$ positive $\vee$ negative = negative
5. For what x is $f(x) = 0$?
 $(-6, 0) (0, 0) (6, 0) (12, 0)$ $x = -6, 0, 6, 12$
6. For what x is $f(x) > 0$?
 $0 < x < 6$ $(-4, 6)$
7. For what x is $f(x) < 0$?
 $-6 < x < 0, 6 < x < 12$ $(-6, 0) \cup (6, 12)$
8. Domain: $D : \{x \mid -6 \leq x \leq 12\}$ $[-6, 12]$
9. Range: $R : \{f(x) \mid -5 \leq f(x) \leq 5\}$ $[-5, 5]$
10. x Intercept: $(-6, 0), (0, 0), (6, 0), (12, 0)$
11. y intercept: $(0, 0)$
12. How many times does the $y = 2$ cross the graph? 2
13. For what x is $f(x) = 4$?
 $(2, 4) (4, 4)$
14. For what x is $f(x) = -5$?
 $(-3, -5) (9, -5)$

Definition 1.9 (Square Brackets: Inclusive). “[” or “]” The endpoint number is included in the interval (corresponds to $\leq \vee \geq$).

Definition 1.10 (Parentheses: Exclusive). “(“ or “)” The endpoint number is not included in the interval (corresponds to $< \vee >$). It is also always used for infinity (∞).

Example 1.17 (Algebraic).

$$f(x) = \frac{x+2}{x-6}$$

1. $f(1) = \frac{(1)+2}{(1)-6} \rightarrow \frac{3}{5} \quad (1, -\frac{3}{5})$

2. $f(x) = 2 \dots$ find x

$$2 = \frac{x+2}{x-6}$$

$$2x - 12 = x + 2$$

$$x - 12 = 2$$

$$x = 14$$

$$(14, 2)$$

3. Domain:

$$x - 6 = 0$$

$$x = 6$$

$$D = \{x \mid x \neq 6\}$$

$$(-\infty, 6) \cup (6, \infty)$$

4. x intercept: $f(x) = 0$

$$0 = \frac{x+2}{x-6}$$

$$x + 2 = 0$$

$$x = -2$$

$$(-2, 0)$$

5. y intercept:

$$f(0) = \frac{(0)+2}{(0)-6} = -\frac{1}{3}$$

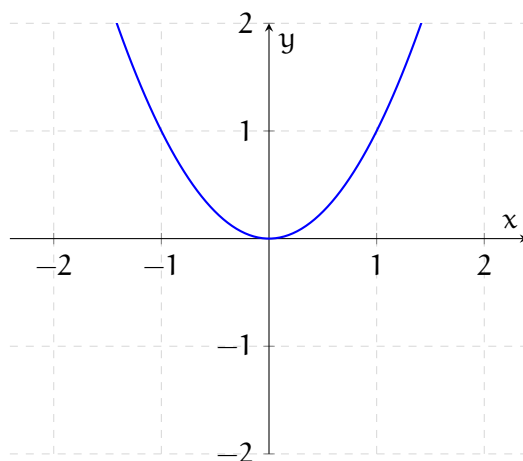
$$(0, -\frac{1}{3})$$

1.7 Properties of Functions: Even or Odd

Definition 1.11 (Even). Even functions are symmetric about y-axes in a graph.

Example 1.18 (Even Function). This is an even function because if we fold it down the y-axis and fold it onto itself, it would be a mirror image perfectly.

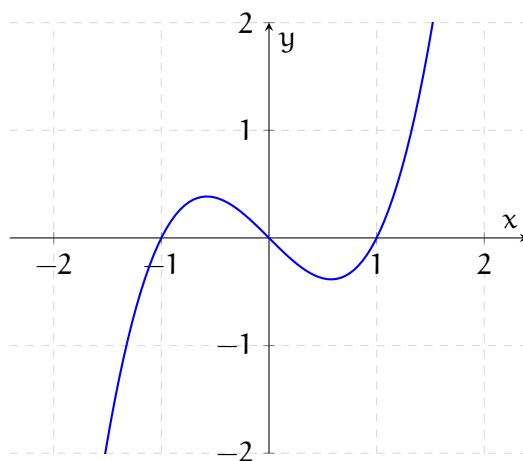
$$f(x) = x^2$$



Definition 1.12 (Odd). Odd functions are symmetric about origins in a graph.

Example 1.19 (Odd Function). This is an odd function because if we rotate the graph 180° about the origin, it will still be the exact same graph.

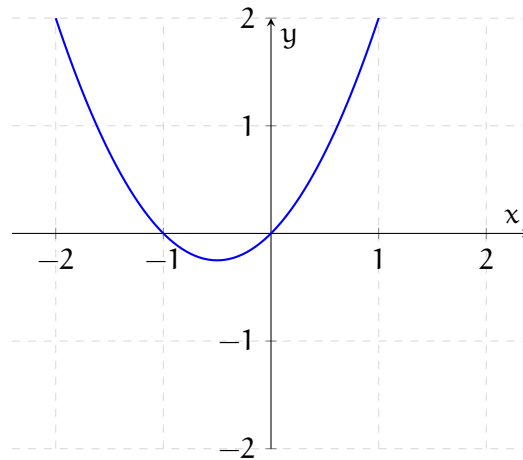
$$f(x) = x^3 - x$$



Note: A lot of graphs are neither even or odd. For example:

Example 1.20 (Neither).

$$f(x) = x^2 + x$$



Definition 1.13 (Algebraic Definition of Even Functions).

$$f(x) = f(-x)$$

Definition 1.14 (Algebraic Definition of Odd Functions).

$$-f(x) = f(-x)$$

Example 1.21 (Algebraic).

- $f(x) = 2x^4 - x^2$
 $f(-x) = 2(-x)^4 - (-x)^2$
 $f(-x) = 2x^4 - x^2$ Even
- $g(x) = \frac{x}{x^2-1}$
 $g(-x) = \frac{(-x)^1}{(-x)^2-1} \rightarrow g(-x) = \frac{-x}{x^2-1}$
 $g(-x) = -\frac{x}{x^2-1}$ Odd
- $h(x) = 3x^3 + 5$
 $h(-x) = 3(-x)^3 + 5 \rightarrow h(-x) = -3x^3 + 5$
 $h(-x) = -(3x^3 - 5)$ Neither
- $F(x) = \frac{x^2+3}{x^2-1}$
 $F(-x) = \frac{(-x)^2+3}{(-x)^2-1}$
 $F(-x) = \frac{x^2+3}{x^2-1}$ Even
- $G(x) = \sqrt{x}$
 $G(-x) = \sqrt{(-x)}$ Neither because it's not a $\mathbb{R}$ number
- $H(x) = \frac{-x^3}{3x^2-9} \rightarrow -\frac{x^3}{3x^2-9}$
 $H(-x) = \frac{-(-x)^3}{3x^2-9}$
 $H(-x) = \frac{x^3}{3x^2-9}$ Odd

1.8 Properties of Functions: Increasing or Decreasing

Definition 1.15 (Increasing).

$$\begin{aligned} &\text{for } x_1 < x_2 \\ &f(x_1) < f(x_2) \end{aligned}$$

From left $\rightarrow$ right in the x -axis, the graph climbs upward.

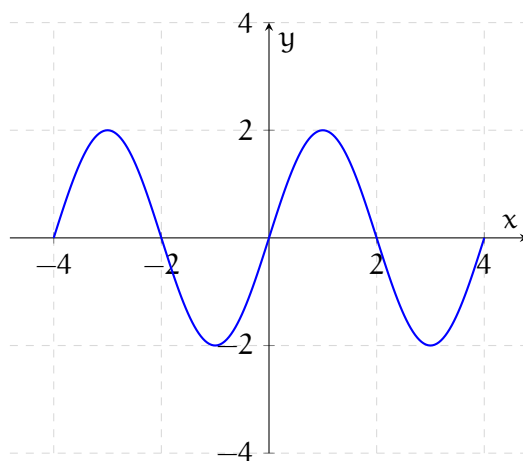
Definition 1.16 (Decreasing).

$$\begin{aligned} &\text{for } x_1 < x_2 \\ &f(x_1) > f(x_2) \end{aligned}$$

From left $\rightarrow$ right in the x -axis, the graph falls downward.

Note: All of this is given by an open intervals of the x -axes.

Example 1.22 (Decreasing and Increasing Function).

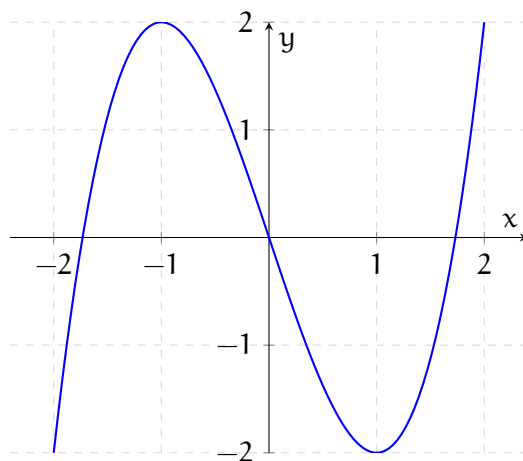


Increasing $[-1, 1] \cup [3, 4]$

Decreasing $[-3, -1] \cup [1, 3]$

Another example is:

$$f(x) = -x^3 - 3 * x$$



Increasing $[-1, 1]$

Decreasing $(-\infty, -1] \cup [1, \infty)$

1.9 Properties of Functions: Extremas

Definition 1.17 (Local/Relative Maximum). The highest point in a specific region or a small interval of a function.

$$f(x) \leq f(c) \dots c \text{ is a local max}$$

Where functions changes from increasing to decreasing.

Definition 1.18 (Local/Relative Minimum). The lowest point in a specific region or a small interval of a function.

$$f(x) \geq f(c) \dots c \text{ is a local min}$$

Where functions changes from decreasing to increasing.

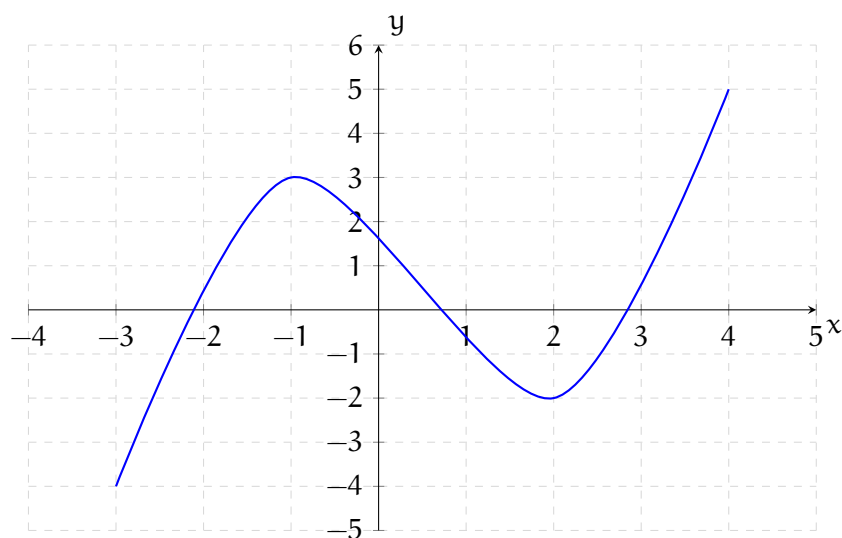
Note: Endpoints are not allowed for Local minimum or maximum. But it is allowed for absolute minimum or maximum.

Definition 1.19 (Absolute Maximum). Highest output on graph.

Definition 1.20 (Absolute Minimum). Lowest output on graph

Note: Every continuous closed interval "[,]" will guarantee we have an absolute max and an absolute min.

Example 1.23 (Graph).



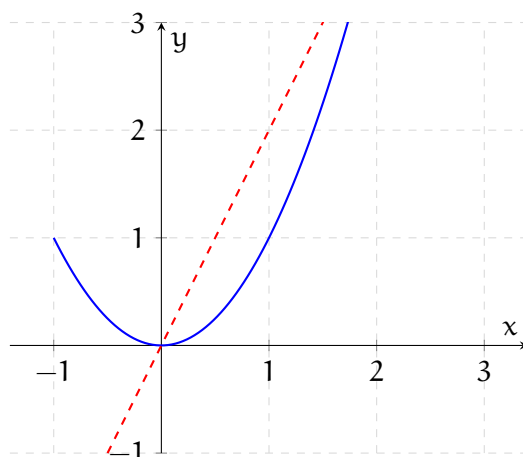
1. Local Minimum: $-2 @ (2, -2)$
2. Local Maximum: $3 @ (-1, 3)$
3. Absolute Minimum: $-4 @ (-3, -4)$
4. Absolute Maximum: $5 @ (4, 5)$

1.10 Average Rate of Change

Definition 1.21 (Average Rate of Change). The slope of the line connecting two points on a curve, known as the *secant line*.

$$\text{Average Rate of Change} = \frac{f(b) - f(a)}{b - a}$$

Example 1.24 (Graph).



$$a = 0, b = 2$$

$$\frac{f(2) - f(0)}{2 - 0} = \frac{4 - 0}{2 - 0} = 2$$

Example 1.25 (Algebraic).

$f(x) = x^2 - 2x$ Find average rate of change

From $x = 3$ to $x = 5$

$$f(3) = (3)^2 - 2(3)$$

$$f(3) = 3$$

$$f(5) = (5)^2 - 2(5)$$

$$f(5) = 15$$

$$(3, 3) \quad (5, 15)$$

$$\text{Average Rate of Change} = \frac{15 - 3}{5 - 3}$$

$$\text{Average Rate of Change} = \frac{12}{2} \rightarrow 6$$

1.11 Piece-wise Functions

Definition 1.22 (Piece-wise Functions). A single function that uses two or more different formulas or rules, with each rule applying to a specific portion of the function's domain.

Example 1.26 (Algebraic).

1.

$$f(x) = \begin{cases} x^3 & \text{if } x < 1 \\ 3x - 2 & \text{if } x \geq 1 \end{cases}$$

(a) $f(-1) = (-1)^3 \rightarrow (-1, -1)$

(b) $f(0) = (0)^3 \rightarrow (0, 0)$

(c) $f(1) = 3(1) - 2 \rightarrow (1, 1)$

2.

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ 2x + 1 & \text{if } x > 0 \end{cases}$$

(a) $f(-2) = (-2)^2 \rightarrow (-2, 4)$

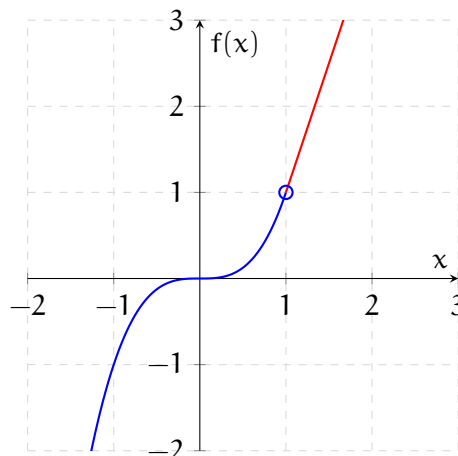
(b) $f(0) = 2 \rightarrow (0, 2)$

(c) $f(2) = 2(2) + 1 \rightarrow (2, 5)$

Example 1.27 (Graph).

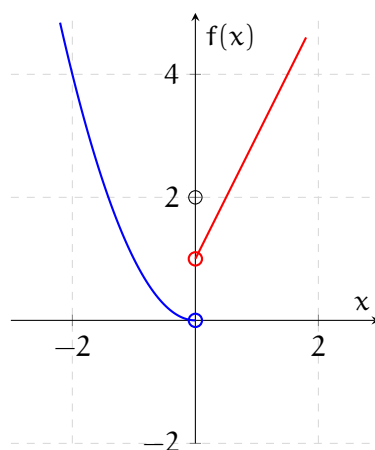
1.

$$f(x) = \begin{cases} x^3 & \text{if } x < 1 \\ 3x - 2 & \text{if } x \geq 1 \end{cases}$$



2.

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ 2x + 1 & \text{if } x > 0 \end{cases}$$

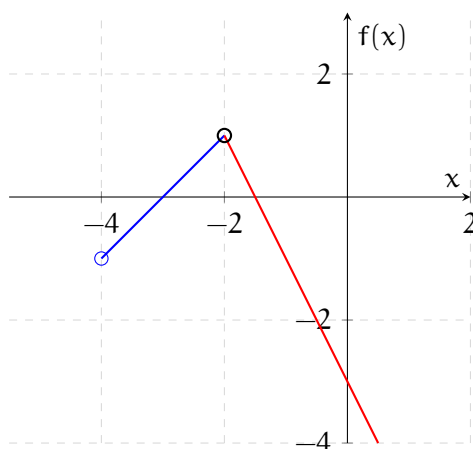


Definition 1.23 (Not Continuous Function). In a piece-wise function, a function is not continuous (*discontinuous*) at a point if the different sub-functions do not connect seamlessly at the transition boundary

Example 1.28 (Not Continuous Function).

$$f(x) = \begin{cases} x + 3 & \text{if } -4 \leq x < -2 \\ -2x - 3 & \text{if } x > -2 \end{cases}$$

Example 1.29 (Graph).



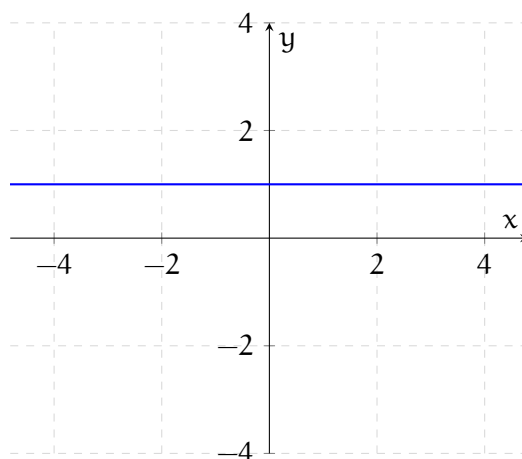
1.12 Important Functions

Example 1.30 (Even). All of the examples have the point $(0,0)$ $(1,1)$ $(-1,-1)$ except for the $f(x) = b$ which is just a constant value and the $f(x) = \sqrt{x}$ function which can't have a negative input ($x \geq 0$).

1.

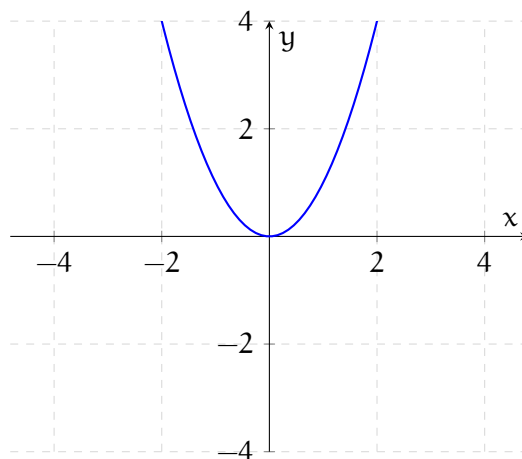
$$f(x) = b$$

For this example $b = 1$



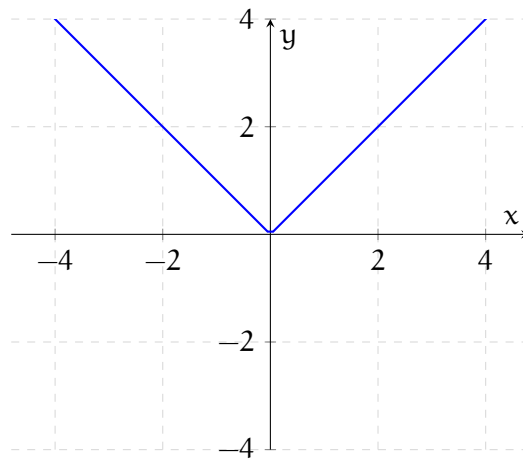
2.

$$f(x) = x^2$$



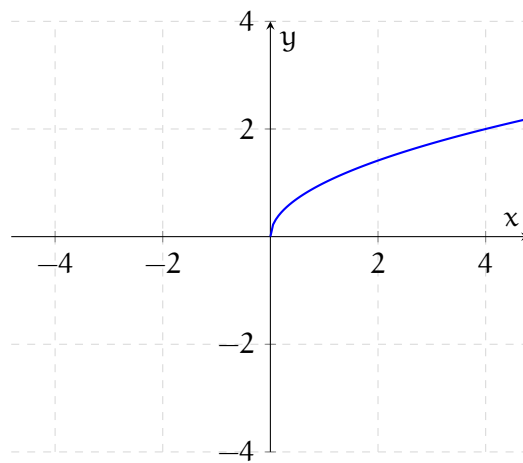
3.

$$f(x) = |x|$$



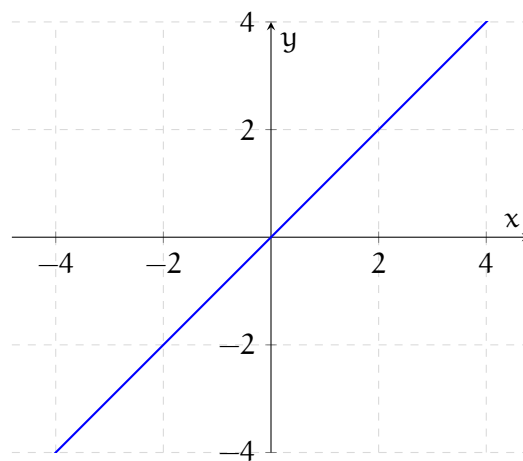
4.

$$f(x) = \sqrt{x}$$

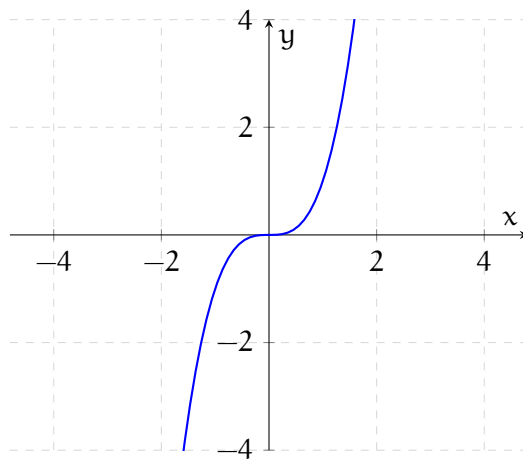


Example 1.31 (Odd). All of the examples have the point $(0,0)$ $(1,1)$ $(-1,-1)$ except for the $f(x) = \frac{1}{x}$ function which doesn't have the $(0,0)$ point.

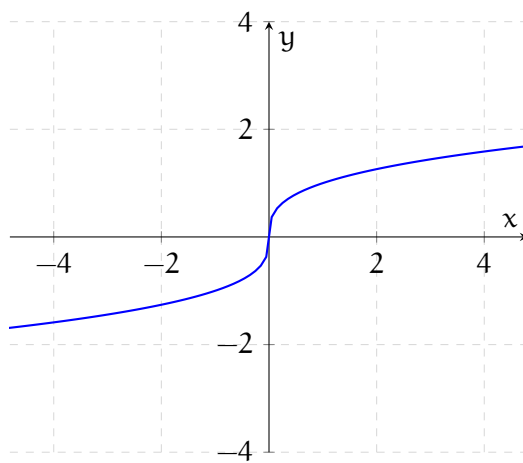
$$f(x) = x$$



$$f(x) = x^3$$



$$f(x) = \sqrt[3]{x}$$



$$f(x) = \frac{1}{x}$$

